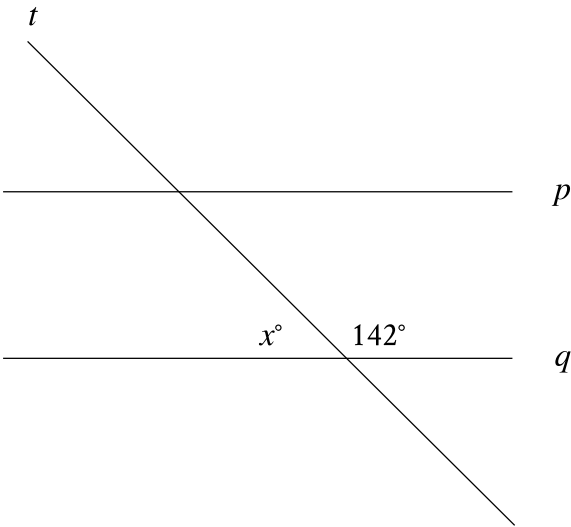


Question ID 3b44439b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 3b44439b



Note: Figure not drawn to scale.

In the figure, line p is parallel to line q , and line t intersects both lines. What is the value of $x + 142$?

- A. 52
- B. 90
- C. 142
- D. 180

ID: 3b44439b Answer

Correct Answer:
D

Rationale

Choice D is correct. In the figure shown, the angle marked x° and the angle marked 142° form a linear pair of angles. If two angles form a linear pair of angles, the sum of the measures of the angles is 180° . Therefore, the value of $x + 142$ is 180.

Choice A is incorrect. This is 90 less than 142, not the sum of x and 142.

Choice B is incorrect and may result from conceptual or calculation errors.

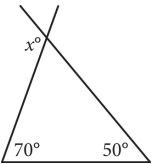
Choice C is incorrect. This is the measure, in degrees, of one of the angles shown.

Question Difficulty:
Easy

Question ID 36200a38

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 36200a38



In the figure above, two sides of a triangle are extended. What is the value of x ?

- A. 110
- B. 120
- C. 130
- D. 140

ID: 36200a38 Answer

Correct Answer:

B

Rationale

Choice B is correct. The sum of the interior angles of a triangle is 180° . The measures of the two interior angles of the given triangle are shown. Therefore, the measure of the third interior angle is $180^\circ - 70^\circ - 50^\circ = 60^\circ$. The angles of measures x° and 60° are supplementary, so their sum is 180° . Therefore, $x = 180 - 60 = 120$.

Choice A is incorrect and may be the result of misinterpreting x° as supplementary to 70° . Choice C is incorrect and may be the result of misinterpreting x° as supplementary to 50° . Choice D is incorrect and may be the result of a calculation error.

Question Difficulty:

Easy

Question ID 42b4493b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 42b4493b

In a right triangle, the measure of one of the acute angles is 51° . What is the measure, in degrees, of the other acute angle?

- A. 6
- B. 39
- C. 49
- D. 51

ID: 42b4493b Answer

Correct Answer:

B

Rationale

Choice B is correct. The sum of the measures of the interior angles of a triangle is **180** degrees. Since the triangle is a right triangle, it has one angle that measures **90** degrees. Therefore, the sum of the measures, in degrees, of the remaining two angles is **180 — 90**, or **90**. It's given that the measure of one of the acute angles in the triangle is **51** degrees. Therefore, the measure, in degrees, of the other acute angle is **90 — 51**, or **39**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the measure, in degrees, of the acute angle whose measure is given.

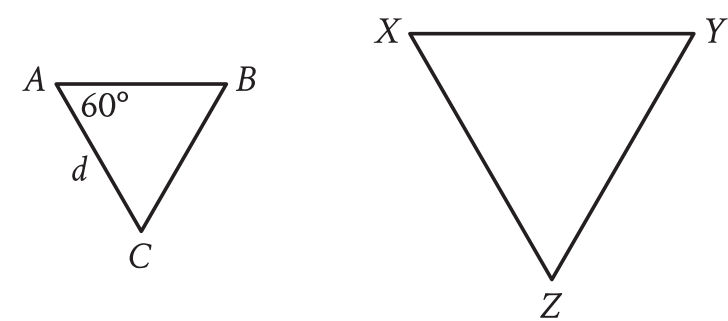
Question Difficulty:

Easy

Question ID e0d2e21a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: e0d2e21a



Note: Figures not drawn to scale.

For the triangles shown, triangle ABC is dilated by a scale factor of 3 to obtain triangle XYZ , where $d = 16$. What is the measure, in degrees, of angle X ?

- A. 20
- B. 57
- C. 60
- D. 63

ID: e0d2e21a Answer

Correct Answer:
C

Rationale
Choice C is correct. It's given that triangle XYZ is obtained by a dilation of triangle ABC . It follows that triangle ABC is similar to triangle XYZ , where A corresponds to X . Since corresponding angles in similar triangles have the same measure and the measure of angle A is 60 degrees, it follows that the measure of angle X is also 60 degrees.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty:
Easy

Question ID 739f1bbc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 739f1bbc

In triangle ABC , $AB = 4,680$ millimeters (mm) and $BC = 4,680 \text{ mm}$. Which statement is sufficient to prove that triangle ABC is equilateral?

- A. $AC = 4,680 \text{ mm}$
- B. $AC = 468 \text{ mm}$
- C. $AC = 46.8 \text{ mm}$
- D. $AC = 4.68 \text{ mm}$

ID: 739f1bbc Answer

Correct Answer:

A

Rationale

Choice A is correct. In an equilateral triangle, all three sides have the same length. It's given that in triangle ABC , $AB = 4,680 \text{ mm}$ and $BC = 4,680 \text{ mm}$. Therefore, if $AC = 4,680 \text{ mm}$, then all three sides of triangle ABC have the same length, so triangle ABC is equilateral. Therefore, $AC = 4,680 \text{ mm}$ is sufficient to prove that triangle ABC is equilateral.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

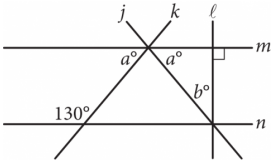
Question Difficulty:

Easy

Question ID 3828f53d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 3828f53d



Note: Figure not drawn to scale.

In the figure above, lines m and n are parallel.
What is the value of b ?

- A. 40
- B. 50
- C. 65
- D. 80

ID: 3828f53d Answer

Correct Answer:
A

Rationale

Choice A is correct. Given that lines m and n are parallel, the angle marked 130° must be supplementary to the leftmost angle marked a° because they are same-side interior angles. Therefore, $130^\circ + a^\circ = 180^\circ$, which yields $a = 50^\circ$. Lines j , k , and m intersect at a right angle, so lines j , k , and m form a right triangle where the two acute angles are a° and b° . The acute angles of a right triangle are complementary, so $a^\circ + b^\circ = 90^\circ$, which yields $50^\circ + b^\circ = 90^\circ$, and $b = 40$.

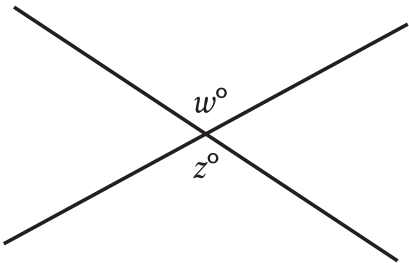
Choice B is incorrect. This is the value of a , not b . Choice C is incorrect and may be the result of dividing 130° by 2. Choice D is incorrect and may be the result of multiplying b by 2.

Question Difficulty:
Easy

Question ID 64d1f49f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 64d1f49f



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $w = 136$, what is the value of z ?

- A. 36
- B. 44
- C. 68
- D. 136

ID: 64d1f49f Answer

Correct Answer:

D

Rationale

Choice D is correct. In the figure shown, the angles with measures w° and z° are vertical angles. Since vertical angles are congruent, $w = z$. Therefore, if $w = 136$, the value of z is **136**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure, in degrees, of an angle that's supplementary, not congruent, to the angle with measure w° .

Choice C is incorrect and may result from conceptual or calculation errors.

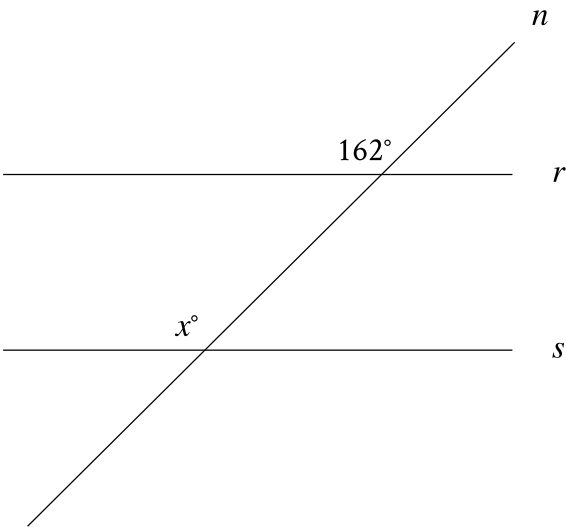
Question Difficulty:

Easy

Question ID 5b918ebb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 5b918ebb



Note: Figure not drawn to scale.

In the figure, line n intersects lines r and s . Line r is parallel to line s . What is the value of x ?

ID: 5b918ebb Answer

Correct Answer:

162

Rationale

The correct answer is **162**. It's given that line r is parallel to line s . Since line n intersects both lines r and s , it's a transversal. The angles in the figure marked as 162° and x° are on the same side of the transversal, where one is an interior angle with line s as a side, and the other is an exterior angle with line r as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that the value of x is **162**.

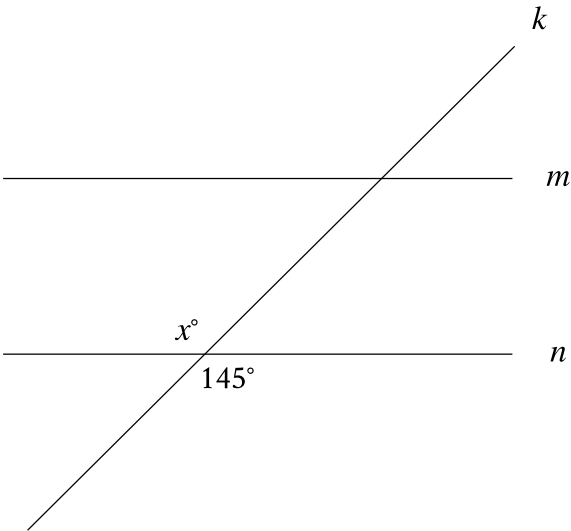
Question Difficulty:

Easy

Question ID 43236565

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 43236565



Note: Figure not drawn to scale.

In the figure, line *m* is parallel to line *n*, and line *k* intersects both lines. Which of the following statements is true?

- A. The value of *x* is less than 145.
- B. The value of *x* is greater than 145.
- C. The value of *x* is equal to 145.
- D. The value of *x* cannot be determined.

ID: 43236565 Answer

Correct Answer:
C

Rationale

Choice C is correct. Vertical angles, or angles that are opposite each other when two lines intersect, are congruent. It’s given that line *k* intersects line *n*. Based on the figure, the angle with measure *x*° and the angle with measure 145° are vertical angles. Therefore, the value of *x* is equal to 145.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Question ID 0bb39de4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 0bb39de4

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 18° . What is the measure of angle F ?

- A. 18°
- B. 72°
- C. 90°
- D. 162°

ID: 0bb39de4 Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that triangle ABC is congruent to triangle DEF . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that angle A corresponds to angle D , and that the measure of angle A is 18° . It's also given that the measures of angles B and E are 90° . Since these angles have equal measure, they are corresponding angles. It follows that angle C corresponds to angle F . Let x° represent the measure of angle C . Since the sum of the measures of the interior angles of a triangle is 180° , it follows that $18^\circ + 90^\circ + x^\circ = 180^\circ$, or $108^\circ + x^\circ = 180^\circ$. Subtracting 108° from both sides of this equation yields $x^\circ = 72^\circ$. Therefore, the measure of angle C is 72° . Since angle C corresponds to angle F , it follows that the measure of angle F is also 72° .

Choice A is incorrect. This is the measure of angle D , not the measure of angle F .

Choice C is incorrect. This is the measure of angle E , not the measure of angle F .

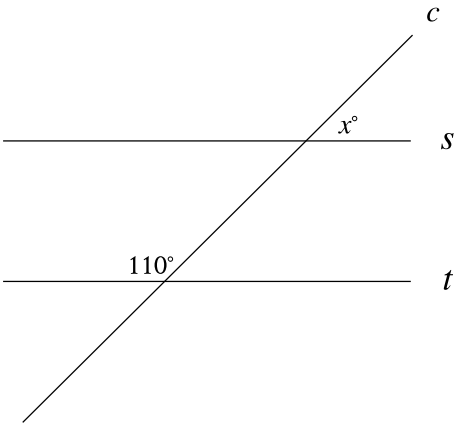
Choice D is incorrect. This is the sum of the measures of angles E and F , not the measure of angle F .

Question Difficulty:
Easy

Question ID cf0d3050

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: cf0d3050



Note: Figure not drawn to scale.

In the figure shown, line *c* intersects parallel lines *s* and *t*. What is the value of *x*?

ID: cf0d3050 Answer

Correct Answer:
70

Rationale

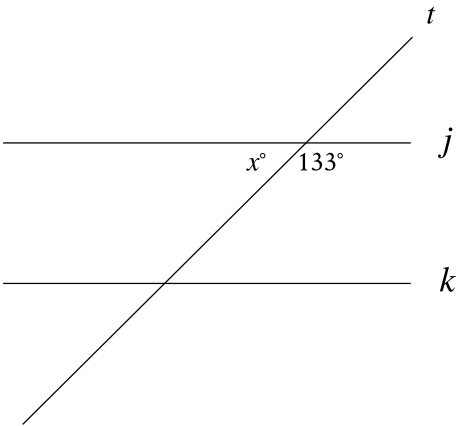
The correct answer is **70**. Based on the figure, the angle with measure **110°** and the angle vertical to the angle with measure *x*° are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure *x*° also has measure *x*°. It's given that lines *s* and *t* are parallel. Therefore, same side interior angles between lines *s* and *t* are supplementary. It follows that ***x* + 110 = 180**. Subtracting **110** from both sides of this equation yields ***x* = 70**.

Question Difficulty:
Easy

Question ID 3b4b5b1e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 3b4b5b1e



Note: Figure not drawn to scale.

In the figure, line *j* is parallel to line *k*. What is the value of *x*?

ID: 3b4b5b1e Answer

Correct Answer:
47

Rationale

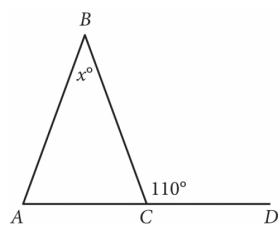
The correct answer is **47**. Based on the figure, the angle with measure x° and the angle with measure 133° together form a straight line. Therefore, these two angles are supplementary, so the sum of their measures is 180° . It follows that $x + 133 = 180$. Subtracting **133** from both sides of this equation yields $x = 47$.

Question Difficulty:
Easy

Question ID 5733ce30

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 5733ce30



In the given figure, $\overline{AC}$ extends to point D . If the measure of $\angle BAC$ is equal to the measure of $\angle BCA$, what is the value of x ?

- A. 110
- B. 70
- C. 55
- D. 40

ID: 5733ce30 Answer

Correct Answer:
D

Rationale

Choice D is correct. Since $\angle BCD$ and $\angle BCA$ form a linear pair of angles, their measures sum to 180° . It's given that the measure of $\angle BCD$ is 110° . Therefore, $110^\circ + \angle BCA = 180^\circ$. Subtracting 110° from both sides of this equation gives the measure of $\angle BCA$ as 70° . It's also given that the measure of $\angle BAC$ is equal to the measure of $\angle BCA$. Thus, the measure of $\angle BAC$ is also 70° . The measures of the interior angles of a triangle sum to 180° . Thus, $70^\circ + 70^\circ + x^\circ = 180^\circ$. Combining like terms on the left-hand side of this equation yields $140^\circ + x^\circ = 180^\circ$. Subtracting 140° from both sides of this equation yields $x^\circ = 40^\circ$, or $x = 40$.

Choice A is incorrect. This is the value of the measure of $\angle BCD$. Choice B is incorrect. This is the value of the measure of each of the other two interior angles, $\angle BCA$ and $\angle BAC$. Choice C is incorrect and may result from an error made when identifying the relationship between the exterior angle of a triangle and the interior angles of the triangle.

Question Difficulty:
Easy

Question ID f1747a6a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: f1747a6a

In triangle ABC , the measure of angle B is 52° and the measure of angle C is 17° . What is the measure of angle A ?

- A. 21°
- B. 35°
- C. 69°
- D. 111°

ID: f1747a6a Answer

Correct Answer:

D

Rationale

Choice D is correct. The sum of the angle measures of a triangle is 180° . Adding the measures of angles B and C gives $52 + 17 = 69^\circ$. Therefore, the measure of angle A is $180 - 69 = 111^\circ$.

Choice A is incorrect and may result from subtracting the sum of the measures of angles B and C from 90° , instead of from 180° .

Choice B is incorrect and may result from subtracting the measure of angle C from the measure of angle B .

Choice C is incorrect and may result from adding the measures of angles B and C but not subtracting the result from 180° .

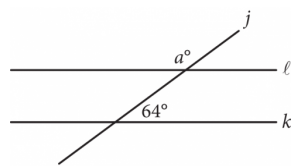
Question Difficulty:

Easy

Question ID 992f4e93

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 992f4e93



Note: Figure not drawn to scale.

In the figure above, lines ℓ and k are parallel.

What is the value of a ?

- A. 26
- B. 64
- C. 116
- D. 154

ID: 992f4e93 Answer

Correct Answer:

C

Rationale

Choice C is correct. Since lines ℓ and k are parallel, corresponding angles formed by the intersection of line j with lines ℓ and k are congruent. Therefore, the angle with measure a° must be the supplement of the angle with measure 64° . The sum of two supplementary angles is 180° , so $a = 180 - 64 = 116$.

Choice A is incorrect and likely results from thinking the angle with measure a° is the complement of the angle with measure 64° . Choice B is incorrect and likely results from thinking the angle with measure a° is congruent to the angle with measure 64° . Choice D is incorrect and likely results from a conceptual or computational error.

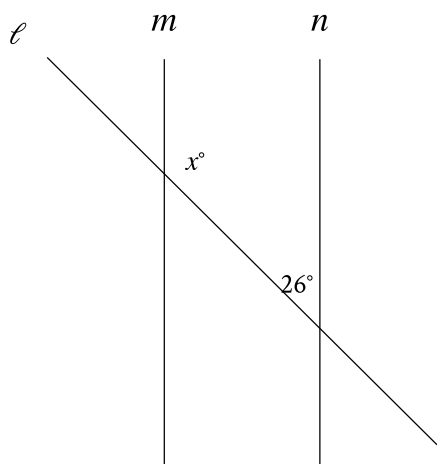
Question Difficulty:

Easy

Question ID afa3c48b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: afa3c48b



Note: Figure not drawn to scale.

In the figure shown, line *m* is parallel to line *n*. What is the value of *x*?

- A. 13
- B. 26
- C. 52
- D. 154

ID: afa3c48b Answer

Correct Answer:

D

Rationale

Choice D is correct. The sum of consecutive interior angles between two parallel lines and on the same side of the transversal is **180** degrees. Since it's given that line *m* is parallel to line *n*, it follows that $x + 26 = 180$. Subtracting **26** from both sides of this equation yields **154**. Therefore, the value of *x* is **154**.

Choice A is incorrect. This is half of the given angle measure.

Choice B is incorrect. This is the value of the given angle measure.

Choice C is incorrect. This is twice the value of the given angle measure.

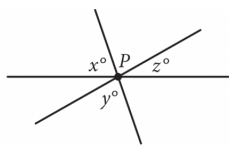
Question Difficulty:

Easy

Question ID 087cdcf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 087cdcf



Note: Figure not drawn to scale.

In the figure, three lines intersect at point P . If $x = 65$ and $y = 75$, what is the value of z ?

- A. 140
- B. 80
- C. 40
- D. 20

ID: 087cdcf Answer

Correct Answer:
C

Rationale

Choice C is correct. The angle that is shown as lying between the y° angle and the z° angle is a vertical angle with the x° angle. Since vertical angles are congruent and $x = 65$, the angle between the y° angle and the z° angle measures 65° . Since the 65° angle, the y° angle, and the z° angle are adjacent and form a straight angle, it follows that the sum of the measures of these three angles is 180° , which is represented by the equation $65^\circ + y^\circ + z^\circ = 180^\circ$. It's given that $y = 75$. Substituting 75 for y yields $65^\circ + 75^\circ + z^\circ = 180^\circ$, which can be rewritten as $140^\circ + z^\circ = 180^\circ$. Subtracting 140° from both sides of this equation yields $z^\circ = 40^\circ$. Therefore, $z = 40$.

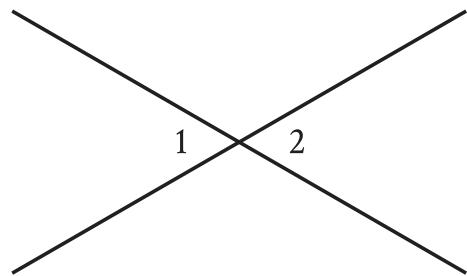
Choice A is incorrect and may result from finding the value of $x + y$ rather than z . Choices B and D are incorrect and may result from conceptual or computational errors.

Question Difficulty:
Easy

Question ID a456f28c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: a456f28c



Note: Figure not drawn to scale.
In the figure, two lines intersect at a point. Angle 1 and angle 2 are vertical angles. The measure of angle 1 is 72° . What is the measure of angle 2?

- A. 72°
- B. 108°
- C. 144°
- D. 288°

ID: a456f28c Answer

Correct Answer:
A

Rationale

Choice A is correct. It’s given that angle 1 and angle 2 are vertical angles, and the measure of angle 1 is 72° . Vertical angles have equal measures. Therefore, the measure of angle 2 is 72° .

Choice B is incorrect. This is the measure of an angle that is supplementary, not congruent, to angle 1.

Choice C is incorrect. This is the sum of the measures of angle 1 and angle 2.

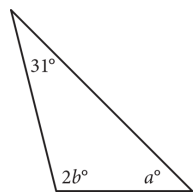
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 410bdbe6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 410bdbe6



In the triangle above, $a = 45$. What is the value of b ?

- A. 52
- B. 59
- C. 76
- D. 104

ID: 410bdbe6 Answer

Correct Answer:
A

Rationale

Choice A is correct. The sum of the measures of the three interior angles of a triangle is 180° . Therefore, $31 + 2b + a = 180$. Since it's given that $a = 45$, it follows that $31 + 2b + 45 = 180$, or $2b = 104$. Dividing both sides of this equation by 2 yields $b = 52$.

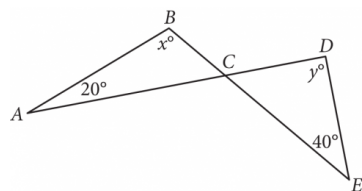
Choice B is incorrect and may result from a calculation error. Choice C is incorrect. This is the value of $a + 31$. Choice D is incorrect. This is the value of $2b$.

Question Difficulty:
Easy

Question ID dfc420b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: dfc420b2



Note: Figure not drawn to scale.

In the figure above, $\overline{AD}$ intersects $\overline{BE}$ at C. If $x = 100$, what is the value of y ?

- A. 100
- B. 90
- C. 80
- D. 60

ID: dfc420b2 Answer

Correct Answer:
C

Rationale

Choice C is correct. It’s given that $x = 100$; therefore, substituting 100 for x in triangle ABC gives two known angle measures for this triangle. The sum of the measures of the interior angles of any triangle equals 180° . Subtracting the two known angle measures of triangle ABC from 180° gives the third angle measure: $180^\circ - 100^\circ - 20^\circ = 60^\circ$. This is the measure of angle BCA. Since vertical angles are congruent, the measure of angle DCE is also 60° . Subtracting the two known angle measures of triangle CDE from 180° gives the third angle measure: $180^\circ - 60^\circ - 40^\circ = 80^\circ$. Therefore, the value of y is 80.

Choice A is incorrect and may result from a calculation error. Choice B is incorrect and may result from classifying angle CDE as a right angle. Choice D is incorrect and may result from finding the measure of angle BCA or DCE instead of the measure of angle CDE.

Question Difficulty:
Easy

Question ID 3563d76d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 3563d76d

At a certain time and day, the Washington Monument in Washington, DC, casts a shadow that is 300 feet long. At the same time, a nearby cherry tree casts a shadow that is 16 feet long. Given that the Washington Monument is approximately 555 feet tall, which of the following is closest to the height, in feet, of the cherry tree?

- A. 10
- B. 20
- C. 30
- D. 35

ID: 3563d76d Answer

Rationale

Choice C is correct. There is a proportional relationship between the height of an object and the length of its shadow. Let c represent the height, in feet, of the cherry tree. The given relationship can be expressed by the proportion $\frac{555}{300} = \frac{c}{16}$. Multiplying both sides of this equation by 16 yields $c = 29.6$. This height is closest to the value given in choice C, 30.

Choices A, B, and D are incorrect and may result from calculation errors.

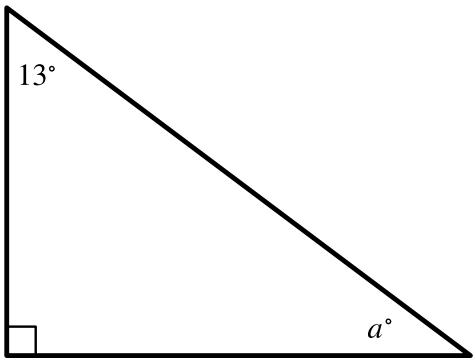
Question Difficulty:

Easy

Question ID 69f4bbdc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 69f4bbdc



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of a ?

- A. 13
- B. 77
- C. 90
- D. 103

ID: 69f4bbdc Answer

Correct Answer:

B

Rationale

Choice B is correct. The triangle shown is a right triangle, where the interior angle shown with a right angle symbol has a measure of 90° . It's shown that the other two interior angles measure 13° and a° . The sum of the measures of the interior angles of a triangle is 180° ; therefore, $90 + 13 + a = 180$. Combining like terms on the left-hand side of this equation yields $103 + a = 180$. Subtracting 103 from both sides of this equation yields $a = 77$.

Choice A is incorrect. This is the measure, in degrees, of the other acute interior angle of the right triangle, not the value of a .

Choice C is incorrect. This is the measure, in degrees, of the right angle of the right triangle, not the value of a .

Choice D is incorrect. This is the sum of the measures, in degrees, of the other two interior angles of the right triangle, not the value of a .

Question Difficulty:

Easy

Question ID b434e103

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: b434e103

In $\triangle RST$, the measure of $\angle R$ is 63° . Which of the following could be the measure, in degrees, of $\angle S$?

- A. 116
- B. 118
- C. 126
- D. 180

ID: b434e103 Answer

Correct Answer:

A

Rationale

Choice A is correct. The sum of the measures of the angles of a triangle is 180° . Therefore, the sum of the measures of $\angle R$, $\angle S$, and $\angle T$ is 180° . It's given that the measure of $\angle R$ is 63° . It follows that the sum of the measures of $\angle S$ and $\angle T$ is $(180 - 63)^\circ$, or 117° . Therefore, the measure of $\angle S$, in degrees, must be less than 117 . Of the given choices, only 116 is less than 117 . Thus, the measure, in degrees, of $\angle S$ could be 116 .

Choice B is incorrect. If the measure of $\angle S$ is 118° , then the sum of the measures of the angles of the triangle is greater than, not equal to, 180° .

Choice C is incorrect. If the measure of $\angle S$ is 126° , then the sum of the measures of the angles of the triangle is greater than, not equal to, 180° .

Choice D is incorrect. This is the sum of the measures of the angles of a triangle, in degrees.

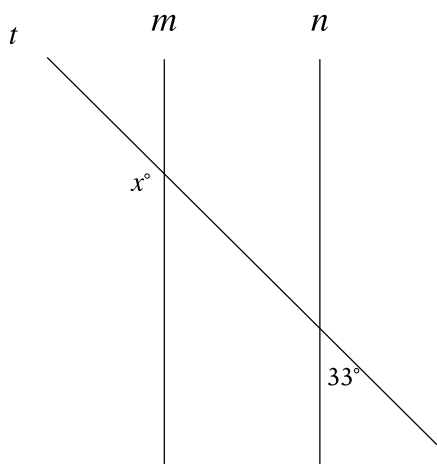
Question Difficulty:

Easy

Question ID 0d3f51dc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 0d3f51dc



Note: Figure not drawn to scale.

In the figure, line *m* is parallel to line *n*, and line *t* intersects both lines. What is the value of *x*?

- A. 33
- B. 57
- C. 123
- D. 147

ID: 0d3f51dc Answer

Correct Answer:

D

Rationale

Choice D is correct. It’s given that line *m* is parallel to line *n*, and line *t* intersects both lines. It follows that line *t* is a transversal. When two lines are parallel and intersected by a transversal, exterior angles on the same side of the transversal are supplementary. Thus, $x + 33 = 180$. Subtracting 33 from both sides of this equation yields $x = 147$. Therefore, the value of *x* is 147.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID f9d40000

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: f9d40000

In $\triangle XYZ$, the measure of $\angle X$ is 23° and the measure of $\angle Y$ is 66° . What is the measure of $\angle Z$?

- A. 43°
- B. 89°
- C. 91°
- D. 179°

ID: f9d40000 Answer

Correct Answer:

C

Rationale

Choice C is correct. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is 180° . It's given that in $\triangle XYZ$, the measure of $\angle X$ is 23° and the measure of $\angle Y$ is 66° . It follows that the measure of $\angle Z$ is $(180 - 23 - 66)^\circ$, or 91° .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the sum of the measures of $\angle X$ and $\angle Y$, not the measure of $\angle Z$.

Choice D is incorrect and may result from conceptual or calculation errors.

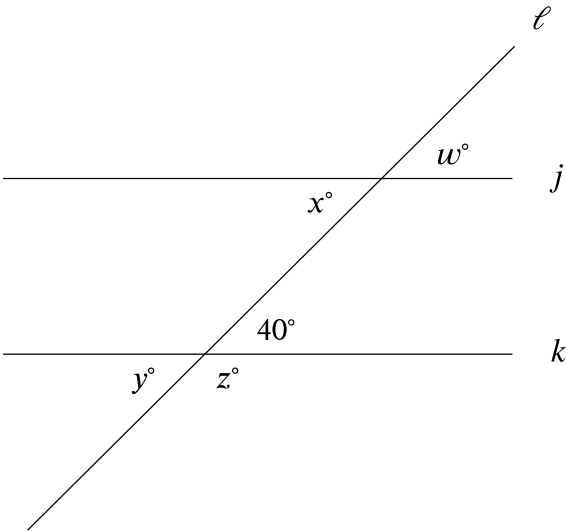
Question Difficulty:

Easy

Question ID 9d078710

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 9d078710



Note: Figure not drawn to scale.

In the figure shown, line ℓ intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

- A. $w = 40$
- B. $x = 140$
- C. $y = 40$
- D. $z = 140$

ID: 9d078710 Answer

Correct Answer:
A

Rationale

Choice A is correct. In the figure shown, lines j and k are parallel if and only if a pair of corresponding angles are congruent. It's given that one angle has a measure of 40° and that the corresponding angle has a measure of w° . Therefore, $w = 40$ is sufficient to prove that lines j and k are parallel.

Choice B is incorrect. The angle measuring x° and the angle measuring 40° are alternate interior angles. Thus, if lines j and k are parallel, x is equal to 40 , not 140 .

Choice C is incorrect. The angle measuring y° and the angle measuring 40° are vertical angles. Thus, $y = 40$, whether lines j and k are parallel or not.

Choice D is incorrect. The angle measuring z° is supplementary to the angle measuring 40° . Thus, $z = 180 - 40$, or $z = 140$, whether lines j and k are parallel or not.

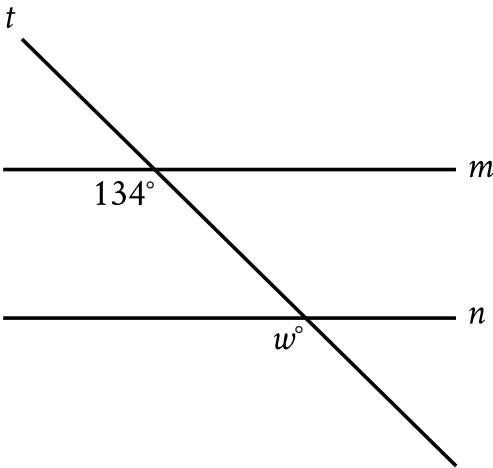
Question Difficulty:

Easy

Question ID c24e1bda

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c24e1bda



Note: Figure not drawn to scale.
In the figure, line m is parallel to line n . What is the value of w ?

- A. 13
- B. 34
- C. 66
- D. 134

ID: c24e1bda Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that lines m and n are parallel. Since line t intersects both lines m and n , it's a transversal. The angles in the figure marked as 134° and w° are on the same side of the transversal, where one is an interior angle with line m as a side, and the other is an exterior angle with line n as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that $w^\circ = 134^\circ$. Therefore, the value of w is 134.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID cbe8ca31

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: cbe8ca31

In $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . What is the measure of $\angle Z$?

- A. 58°
- B. 74°
- C. 122°
- D. 212°

ID: cbe8ca31 Answer

Correct Answer:

A

Rationale

Choice A is correct. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is 180° . It's given that in $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . It follows that the measure of $\angle Z$ is $(180 - 24 - 98)^\circ$, or 58° .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures of $\angle X$ and $\angle Y$, not the measure of $\angle Z$.

Choice D is incorrect and may result from conceptual or calculation errors.

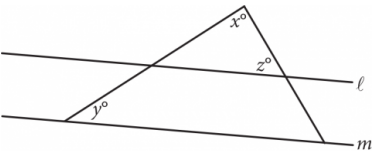
Question Difficulty:

Easy

Question ID a6dbad6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: a6dbad6b



Note: Figure not drawn to scale.

In the figure above, lines ℓ and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

- A. 120
- B. 100
- C. 90
- D. 80

ID: a6dbad6b Answer

Correct Answer:
B

Rationale

Choice B is correct. Let the measure of the third angle in the smaller triangle be a° . Since lines ℓ and m are parallel and cut by transversals, it follows that the corresponding angles formed are congruent. So $a^\circ = y^\circ = 20^\circ$. The sum of the measures of the interior angles of a triangle is 180° , which for the interior angles in the smaller triangle yields $a + x + z = 180$. Given that $z = 60$ and $a = 20$, it follows that $20 + x + 60 = 180$. Solving for x gives $x = 180 - 60 - 20$, or $x = 100$.

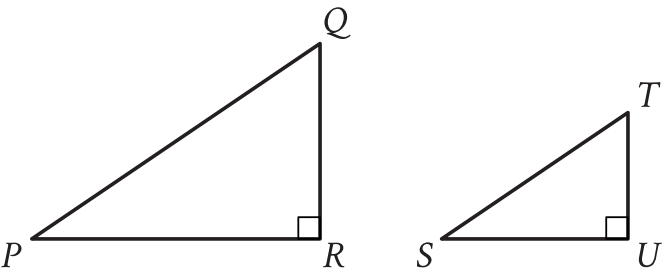
Choice A is incorrect and may result from incorrectly assuming that angles $x + z = 180$. Choice C is incorrect and may result from incorrectly assuming that the smaller triangle is a right triangle, with x as the right angle. Choice D is incorrect and may result from a misunderstanding of the exterior angle theorem and incorrectly assuming that $x = y + z$.

Question Difficulty:
Easy

Question ID d5f349b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: d5f349b7



Note: Figures not drawn to scale.

Right triangles PQR and STU are similar, where P corresponds to S . If the measure of angle Q is 18° , what is the measure of angle S ?

- A. 18°
- B. 72°
- C. 82°
- D. 162°

ID: d5f349b7 Answer

Correct Answer:

B

Rationale

Choice B is correct. In similar triangles, corresponding angles are congruent. It's given that right triangles PQR and STU are similar, where angle P corresponds to angle S . It follows that angle P is congruent to angle S . In the triangles shown, angle R and angle U are both marked as right angles, so angle R and angle U are corresponding angles. It follows that angle Q and angle T are corresponding angles, and thus, angle Q is congruent to angle T . It's given that the measure of angle Q is 18° , so the measure of angle T is also 18° . Angle U is a right angle, so the measure of angle U is 90° . The sum of the measures of the interior angles of a triangle is 180° . Thus, the sum of the measures of the interior angles of triangle STU is 180 degrees. Let s represent the measure, in degrees, of angle S . It follows that $s + 18 + 90 = 180$, or $s + 108 = 180$. Subtracting 108 from both sides of this equation yields $s = 72$. Therefore, if the measure of angle Q is 18 degrees, then the measure of angle S is 72 degrees.

Choice A is incorrect. This is the measure of angle T .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the sum of the measures of angle S and angle U .

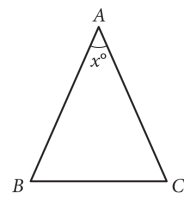
Question Difficulty:

Easy

Question ID c8d60e48

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c8d60e48



In the given triangle, $AB = AC$ and $\angle ABC$ has a measure of 67° . What is the value of x ?

- A. 36
- B. 46
- C. 58
- D. 70

ID: c8d60e48 Answer

Correct Answer:

B

Rationale

Choice B is correct. Since $AB = AC$, the measures of their corresponding angles, $\angle ABC$ and $\angle ACB$, are equal. Since $\angle ABC$ has a measure of 67° , the measure of $\angle ACB$ is also 67° . Since the sum of the measures of the interior angles in a triangle is 180° , it follows that $67 + 67 + x = 180$, or $134 + x = 180$. Subtracting by 134 on both sides of this equation yields $x = 46$.

Choices A, C, and D are incorrect and may result from calculation errors.

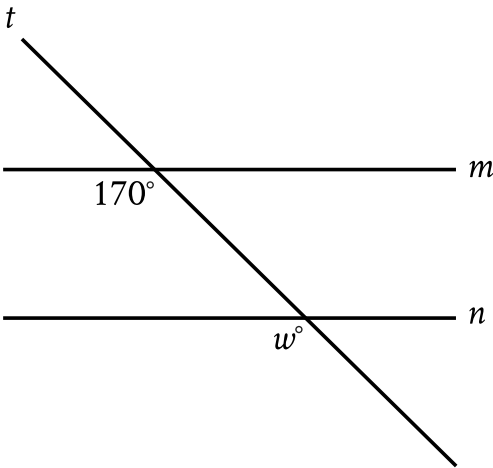
Question Difficulty:

Easy

Question ID 5207e508

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 5207e508



Note: Figure not drawn to scale.
In the figure, line m is parallel to line n . What is the value of w ?

- A. 17
- B. 30
- C. 70
- D. 170

ID: 5207e508 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that lines m and n are parallel. Since line t intersects both lines m and n , it's a transversal. The angles in the figure marked as 170° and w° are on the same side of the transversal, where one is an interior angle with line m as a side, and the other is an exterior angle with line n as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that $w^\circ = 170^\circ$. Therefore, the value of w is 170.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 9912e19f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 9912e19f

Triangles EFG and JKL are congruent, where E , F , and G correspond to J , K , and L , respectively. The measure of angle E is 45° and the measure of angle F is 20° . What is the measure of angle J ?

- A. 20°
- B. 45°
- C. 135°
- D. 160°

ID: 9912e19f Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that triangles EFG and JKL are congruent such that angle E corresponds to angle J . Corresponding angles of congruent triangles are congruent, so angle E and angle J must be congruent. Therefore, if the measure of angle E is 45° , then the measure of angle J is also 45° .

Choice A is incorrect. This is the measure of angle K , not angle J .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy